
Linear Algebra I-II

Theorems & Lemmas

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1 Vector Spaces

1.1 Vector Spaces and Linear Subspaces

Definition 1.1: Vector Space

A **vector space** over a field K is a set V , endowed with two operations

$$\begin{aligned} + : V \times V &\rightarrow V, & (v_1, v_2) &\mapsto v_1 + v_2, \\ \cdot : K \times V &\rightarrow V, & (\alpha, v) &\mapsto \alpha \cdot v \end{aligned}$$

s.t. the following conditions (axioms) hold:

- (V1) $\forall v_1, v_2, v_3 \in V : v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$,
- (V2) $\exists$ an element $0 \in V : 0 + v = v \quad \forall v \in V$,
- (V3) $\forall v \in V \exists v' \in V : v + v' = 0$,
- (V4) $\forall v_1, v_2 \in V : v_1 + v_2 = v_2 + v_1$,
- (V5) $\forall a, b \in K \forall v \in V : a \cdot (b \cdot v) = (a \cdot b) \cdot v$,
- (V6) $\forall v \in V : 1 \cdot v = v$,
- (V7) $\forall a \in K \forall v_1, v_2 \in V : a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$,
- (V8) $\forall a, b \in K \forall v \in V : (a + b) \cdot v = a \cdot v + b \cdot v$.

Lemma 1.2: Properties of Vector Spaces

Let V be a vector space over a field K . Then:

- (a) The 0 vector from (V2) is unique. To avoid confusion (e.g. with $0 \in K$) we will sometimes denote it by 0_V .
- (b) The element v' (corresponding to v) from (V3) is unique. We'll denote it by $-v$. We'll also write $w - v := w + (-v)$ for all $w \in V$,
- (c) $\forall v \in V : 0 \cdot v = 0$,
- (d) $\forall a \in K : a \cdot 0 = 0$,
- (e) $\forall v \in V : -1 \cdot v = -v$,
- (f) $\forall v \in V : -(-v) = v$,
- (g) If $a \cdot v = 0$ for some $a \in K, v \in V$, then either $a = 0$ or $v = 0$,
- (h) Associativity for $+$ holds also for some of n elements. $v_1 + \dots + v_n$ makes sense no matter how we put the brackets $v_1 + (v_2 + (\dots + (v_{n-1} + v_n)))$ or $(v_1 + v_2) + (\dots)$ etc.

Example: The Coordinate Space K^n

Let K be a field and $n \in \mathbb{N}$.

$$K^n = \{(a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n\}.$$

K^n is a vector space over K if we endow it with the following operations:

- Let $v = (a_1, \dots, a_n), w = (b_1, \dots, b_n) \in K^n$.

$$v + w = (a_1, \dots, a_n) + (b_1, \dots, b_n) := (a_1 + b_1, \dots, a_n + b_n),$$

- For $a \in K$

$$a \cdot v = a \cdot (a_1, \dots, a_n) := (a \cdot a_1, \dots, a \cdot a_n)$$

- $0 := (0, \dots, 0) \in K^n$

With these operations K^n becomes a vector space over K .

Definition 1.3: Linear Subspace

Let V be a vector space over K . A subset $W \subseteq V$ is called a **linear subspace** of V (or just subspace of V), if the following holds

(LSS1) $W \neq \emptyset$,

(LSS2) $\forall w_1, w_2 \in W : w_1 + w_2 \in W$,

(LSS3) $\forall a \in K, w \in W : a \cdot w \in W$.

Lemma 1.4: Linear Subspace is a Vector Space

Let V be a vector space over K and $W \subseteq V$ a subspace. Then W is a vector space on its own, when endowed with the operations from V .

Lemma 1.5: Criterion for a Linear Subspace

Let V be a vector space over K , and $W \subseteq V$ a subset. Then W is a linear subspace of V if and only if the following holds:

(1) $0_V \in W$,

(2) $\forall a_1, a_2 \in K, w_1, w_2 \in W : a_1 w_1 + a_2 w_2 \in W$.

Example: Spaces of functions

Fix a field K and a non-empty set S . Define

$$K^S = \{f : S \rightarrow K \mid f \text{ is a function}\}.$$

Define $+$ and $\cdot$ on K^S as follows. Let $f, g \in K^S$, $a \in K$. Define

$$\begin{aligned}(f + g) : S &\rightarrow K, & S \ni x &\mapsto f(x) + g(x), \\ (a \cdot f) : S &\rightarrow K, & S \ni x &\mapsto a \cdot f(x).\end{aligned}$$

Now Let $S = [0, 1]$ and take $K = \mathbb{R}$. Define $C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0, 1]\} \subseteq \mathbb{R}^S$. Then $C(S) \subseteq \mathbb{R}^S$ is a linear subspace.

1.2 Span of Sets

Definition 1.6: Span

Let $S \subseteq V$ be a subset. We define

$$\text{Sp}(S) := \bigcap_{W \in \mathcal{N}} W,$$

where

$$\mathcal{N} := \{W \mid W \text{ is a subspace of } V \text{ and } S \subseteq W\},$$

i.e., the collection of all subspaces $W \subseteq V$ that contain S . $\text{Sp}(S)$ is called the span of S (in V).

Lemma 1.7: Properties of the Span

- (1) $\text{Sp}(S) \subseteq V$ is a linear subspace.
- (2) If $W \subseteq V$ is a linear subspace and $S \subseteq W$, then $\text{Sp}(S) \subseteq W$.

In other words, $\text{Sp}(S)$ is a linear subspace of V , and moreover among those subspaces that contain S , $\text{Sp}(S)$ is the smallest one.

Definition 1.8: Linear Combination

Let $n \in \mathbb{N}$, $a_1, \dots, a_n \in K$, $v_1, \dots, v_n \in V$. A **linear combination** of the vectors $v_1, \dots, v_n$ with coefficients $a_1, \dots, a_n$ is the vector

$$a_1 v_1 + \dots + a_n v_n = \sum_{i=1}^n a_i v_i \in V.$$

It is important to note that in this course consider only finitely many elements in the sums of linear combinations.

Lemma 1.9: Alternate Definition of Span

Let $\emptyset \neq S \subseteq V$. Then

$$\text{Sp}(S) = \{a_1 v_1 + \dots + a_n v_n \mid n \in \mathbb{N}, a_i \in K, v_i \in S \quad \forall 1 \leq i \leq n\}.$$

Remark 1.10. The set S can be finite or infinite. Regardless of that, each linear combination uses only finite numbers of elements from S .

Let $n \in \mathbb{N}$, $v_1, \dots, v_n \in V$. We'll write

$$\text{Sp}(v_1, \dots, v_n) := \text{Sp}(\{v_1, \dots, v_n\}).$$

Remark 1.11. *It obviously holds that*

$$\text{Sp}(v_1, \dots, v_n) = \left\{ \sum_{i=1}^n \alpha_i v_i \mid \alpha_i \in K \quad \forall 1 \leq i \leq n \right\}.$$

Remark 1.12. *It follows from the definition of span that*

$$\text{Sp}(\emptyset) = \{0_V\}.$$

Definition 1.13: Generating Set

Let V be a vect. sp. over K and $S \subseteq V$. We say that S **generates** (or **spans**) V if $\text{Sp}(S) = V$. S is called a generating set of V if $\text{Sp}(S) = W$, where $W \subseteq V$ is a subspace of V . We say that S **generates** (or **spans**) W .

Definition 1.14: Finite Dimensional

A vector space V is called **finite-dimensional** if $\exists$ a **finite** set $S \subseteq V$ such that $\text{Sp}(S) = V$. Otherwise V is called **infinite-dimensional**.

1.3 Bases of Vector Spaces

1.3.1 Linear Independence

Let V be a vector space over K .

Definition 1.15: Linear Independence

Let $v_1, \dots, v_n$ be a list of n vectors in V . We say that $v_1, \dots, v_n$ is **linearly independent** if the only linear combination of $v_1, \dots, v_n$ that gives $0 \in V$ is the trivial one. In other words,

$$\forall a_1, \dots, a_n \in K : a_1 v_1 + \dots + a_n v_n = 0 \implies a_1 = 0, \dots, a_n = 0.$$

By definition, we say that the empty list of vectors $\emptyset$ is linearly independent. We say that a list of vectors is linearly dependent if, the list is not linearly independent.

Remark 1.16. (1) 0 can always be written as a linear combination of any finite list of vectors $0v_1 + \dots + 0v_n = 0$, i.e., the trivial linear combination.

(2) If the list $v_1, \dots, v_n$ has two vectors that are the same, say $v_k = v_l$ for $1 \leq k < l \leq n$, then $v_1, \dots, v_n$ is **not** linearly independent. Indeed, $v_k + (-1)v_l = 0$.

(3) The order of the vectors in the list $v_1, \dots, v_n$ has no effect on linear independence. So, if there are no repetitions in $v_1, \dots, v_n$ we can talk about linear independence of the set $\{v_1, \dots, v_n\}$.

Definition 1.17: Linear Independence in the Context of Families

Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a family of vectors in V , indexed by a set I . We say that $\mathcal{F}$ is linearly independent if $\forall n \in \mathbb{N}$ and any sequence of distinct indices $i_1, \dots, i_n \in I$, the list of vectors $v_{i_1}, \dots, v_{i_n}$ is linearly independent. Of course, if in $\mathcal{F}$ there are any repetitions of vectors, i.e., if $\exists i, j, i \neq j$, s.t. $v_i = v_j$, then $\mathcal{F}$ cannot be linearly independent.

Lemma 1.18: Subsets of Linearly Independent Sets are Linearly Independent

If $\emptyset \neq S \subseteq V$ is linearly independent. Then every non-empty subset $S' \subseteq S$ is also linearly independent.